

Rock de Vocht

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overview

- Experience and skills cover a broad spectrum of IT: data analysis, infrastructure, CI/CD, AI and machine learning, big-data, software development, operating system expertise, IP creation, and mentoring to ensure team success
- Master of Science in Computer Science
- Technical thought leader (creating and sharing with colleagues)
- Natural leader of development teams (works well in groups and teams)

Skills

Recent formal mid-career education (online via Coursera) **2013 - 2014**
(see Education section below)

- Machine Learning (Stanford University, 2013)
 - Writing music like Mozart (Singapore School of Music, 2014)
-

experience

SimSage (UK) Ltd

CSO / CTO role

Cambridge, UK Jan 2020 -

Created a Graph-based Enterprise Semantic Search Engine with workflow and natural language processing. Used for context-engineering for LLMs. Provide a sophisticated way to access/retrieve Enterprise data using Natural Language Processing techniques.

- Created & architected high performance big-data system for millions of documents
- Dev Technology stack: Kotlin (JVM), React js + Typescript, C, Java, Dotnet framework, Linux
- Infrastructure: Ubuntu server (Linux), Kubernetes, Docker
- Created ~ 30 connectors for systems like Google drive, Dropbox, Box, OneDrive, MS Exchange, MS Sharepoint, Egnyte, MS fileshares (Samba), etc.
- Document conversion of ~ 300 different document formats (e.g., pdf, docx, doc, pptx, xlsx...)
- Created language graphs for languages such as English, French, Spanish, Italian, German and Dutch (European market)
- Created and maintained large code base, CI/CD (3,200+ unit tests, ~250,000 lines of code)
- Integration (context engineering) with Gemini, Ollama (local), OpenAI, CoPilot, AWS Bedrock.

MBIE, NZ Immigration

Developer (contract)

Wellington May 2018 – Dec 2019

Visa Assessment Tool (VAT) development, and helping out the Risk Analysis Project (RAP)

- C# development on Azure (dotnet core) on Linux
- Python for data manipulation and neural networks for risk analysis
- using Machine Learning for data quality analysis
- NLP processing of 170M unstructured text records for signal analysis (getting statistically-significant values out of unstructured information)

Booktrack.com

NLP Specialist (2 years contract/permanent)

Wellington/Auckland March 2016 – May 2018

Artificial Intelligence and natural language lead

- Liaising/advising the Booktrack CTO on development strategies, natural language processing and machine learning
- Kubernetes and docker alongside development in C#, Java, and Python, Google cloud and Azure.
- R&D neural networks, machine learning, natural language processing as related to Booktrack's core ip. This includes emotional text analysis, and understanding context in small sections of text.
- Automatic Speech Recognition (ASR) system design for Booktrack to enable automatic transcripts from spoken sounds of various accents to text.
- Develop a series of products related to processing unstructured data and making connections between text and audio.
- Modelling of data-sets into vectors for determining statistical anomalies and use of data in machine learning and other algorithms that solve difficult data matching problems.

Verifone

Senior Developer (contract)

Wellington January 2015 - March 2016

Senior developer in the BAU team. Environment support, bug fixing, troubleshooting. Part of the Payment Gateway team.

- Analysing problems, bugs and fixing them (DevOps)
- Responsible for solving complex issues referred to level-three support

SYL Semantics

Chief Scientist & co-founder

Wellington 2008 – 2015 (January)

Managing software architecture, technologies, technical roadmap and leading the development

team. Created and developed an innovative natural language based semantic search engine for Enterprise Software Solutions. This system is now in use at several New Zealand government departments and is highly regarded. (including MOT, ESR, NZTA, MPI)

- Mentoring developers
- Developing technologies and practical solutions to difficult problems
- Research and Development
- on-site customer implementations
- Developed SYL intellectual property
- Wrote integration software for many external systems (e.g. MS Sharepoint, MS CRM, MS Exchange, IBM WCM, Facebook API, Twitter API)

Datacom Services and Integration

Senior Architect Microsoft Team

Wellington – 2004 – 2008

Developing solutions for Datacom's customers. Mentoring and assisting developers, writing software. pre-sales with technical assistance.

- Pre-sales technical assistance
- Technical writing
- Solution Architecture
- Internal mentoring and education
- Part of the internal Datacom architecture solution group
- Team lead, developer

Binary-star.com

Lead Developer

Devonport – 2002 – 2004

XBOX and PC Video Games development at startup Binary-star. A games development company.

- Xbox and PC C++ developer
- Research and Development into skeletal animation and skin deformation
- Setup development pipeline to help artists ingrate their workflows into development work-streams
- Setup source control systems

Smartsims.com

Senior Developer

Meadowbank – 2000 – 2002

C++ development on a massive online financial market simulator for game-play. Member of the model builder team.

- Research and Development of computer language modelling and design techniques

- Software design and development of a model builder for building financial models in 2- and 3-dimensions

Datacom Systems

Senior Developer and Project Manager

Auckland city – 1999 – 2000

Various internet development. Classic ASP, Java, and generic Object-Oriented design and solution implementation.

Terabyte Interactive

Senior Developer

Newmarket – 1997 – 1999

Senior Analyst, technical lead. C++ development. Designed and created successful proto-type for original Virtual Spectator. Led first Virtual Spectator team to produce Virtual Spectator 1.0. Many other projects. Virtual Spectator was awarded „best technical design” by the Interactive Media Group.

- Research and Development of Virtual Spectator
- Designed and Implemented Wises Maps road-map DVD
- Designed and Implemented Universal Business Directories (UBD) interactive DVD

University of Auckland

System Administrator, Commerce Support Unit (CSU)

Auckland city – 1996 – 1997

Student system administrator. In charge of several computer labs.

- Hardware and desktop support
- Windows NT 3.51 / 4.0 System Administrator
- Student Computer laboratory supervisor
- Implemented University Registrar student account integration with Windows NT users
- Managed a network of 300 computers with 8,000 users (Students)

skills

IP creation, technical thought leadership, Artificial Intelligence, Natural language processing and understanding, Data modelling, Micro-architectures, Google cloud, n-tier architectures, Model Driven Architecture (MDA), Reactive frameworks, Kubernetes, Docker, Object Oriented Programming, Functional Programming and all aspects of the software development life cycle (SDLC). Strong Development / Operations background (DevOps). Mentoring and troubleshooting. Big-data (Ignite, Cassandra, ScyllaDB, Hadoop, Hazelcast). Languages: Kotlin, Java, Go, Python, C#, Javascript, React web, C++. Operating Systems: RedHat/Centos, Ubuntu, various Windows.

natural
languages

English, Dutch, German (proficient), French (proficient),
Russian (learning)

hobbies

Composer of Orchestral Music
Piano performance (grade 8)
Wing Tsun Kung Fu (black belt)

education

Master of Science in Computer Science, University of Auckland

Thesis in Artificial Intelligence, Natural Language Understanding (NLU) and
human computer interaction (HCI).

**Bachelor of Science in Computer Science with honours, University of
Auckland**

Awarded senior prize in Computer Science

New Zealand School of Music (NZSM)

Victoria University, Wellington – (part time)

Completed Stage 1 Music Composition